Adaptive Re-ranking for Efficient Visual Place Recognition

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Abstract

Visual Place Recognition (VPR) is a process for determining a location from an image by comparing it against a geotagged reference database. Most contemporary VPR systems operate through a two-stage pipeline: initial global retrieval, followed by optional geometric refinement. Although refinement with local features significantly enhances accuracy, particularly in challenging domains such as night-time conditions or occlusions, it incurs high computational cost and introduces significant latency.

Our work proposes an adaptive re-ranking strategy to better balance this efficiency-accuracy trade-off. Our method does not apply refinement universally, but instead triggers it selectively for "hard" queries where the initial global retrieval is likely unreliable, using the number of geometric inliers as a discriminative metric. We evaluate threshold-based and logistic regression-based triggering mechanisms, employing the NetVLAD and MegaLoc models and SuperGlue and LoFTR image matching models on the SF-XS, Tokyo-XS, and SVOX datasets. With our method, on the Tokyo-XS dataset, MegLoc+SuperGlue achieve 95.24% Recall@1, comparable to (or slightly outperforming) the full re-ranking baseline at 94.92%, while applying local refinement to only 5.4% of queries. Similar trends hold across other datasets: we attain identical Recall@1/5/10/20 levels on SF-XS, SVOX_Night, and SVOX_Sun, yet trigger re-ranking for just 7.9%, 3.6%, and 1.5% of queries, respectively. These results highlight the effectiveness of selective refinement based on geometric inlier counts, enabling significant efficiency gains without compromising accuracy. The implementation code is available at: https://github.com/leilaalizadeh/VPR_Project

1. Introduction

Visual Place Recognition (VPR) is the task of recognizing the location where an image was taken given only its visual content. It serves as a building block of fundamental importance for applications in autonomous driving, augmented reality, robotics, and simultaneous localization and mapping. The problem is commonly treated as an image retrieval task, where a summary description of a query image is computed and compared against a database of images with known locations.

Typically, a VPR pipeline involves computing descriptors for both the query and the database images. A similarity search is then conducted over the database using a distance metric, often implemented with a K-Nearest Neighbors algorithm. This algorithm identifies the K most similar images to the query based on their mathematical representations. The location is then predicted by selecting the most similar database descriptor; however, additional aggregation techniques may be used to refine the prediction.

The main challenge in VPR is recognizing the same place despite changes in illumination, seasons, or viewpoint. Current state-of-the-art methods struggle when queries are taken in significantly different distributions than the database, such as night-time images or scenes with substantial occlusions from cars and people. To improve performance, many pipelines include an optional refinement step, which seeks to re-rank the nearest neighbors by utilizing local image features. These methods, such as SuperGlue or LoFTR, compare matched keypoints to survive a geometric post-processing check.

While local feature refinement substantially boosts accuracy, it introduces a significant performance-efficiency trade-off. The re-ranking process is computationally expensive and time-consuming, as it requires extracting and matching hundreds of local descriptors for every query. In safety-critical scenarios like autonomous driving, this latency can be prohibitive. Furthermore, current VPR systems often operate deterministically, providing no indication of uncertainty or confidence in their predictions. This lack of reliability is a major hurdle for real-world deployment, as incorrect predictions could lead to risky decisions or environmental damage.

The goal of this project is to investigate and improve the reliability and efficiency of the VPR pipeline. Specifically, we explore the concept of adaptive re-ranking. Instead of blindly applying refinement to every query, we propose a strategy where the system only utilizes the expensive image

matching stage when a query is presumed "hard". By defining query difficulty based on the number of inliers between the query and the top-retrieved image, we can selectively trigger refinement. This approach aims to maintain high localization performance while achieving substantial cost and time savings, moving toward VPR solutions that are both effective and computationally sustainable.

2. Related Work

Visual Place Recognition (VPR) is commonly formulated as an image retrieval problem. Given a query image, the goal is to localize it by retrieving visually similar, geo-referenced images from a database, typically using robust global image descriptors.

2.1. VPR Models

Early deep learning approaches, such as NetVLAD [2], established the foundation for end-to-end global descriptor learning by introducing trainable aggregation layers together with weakly supervised ranking losses. Building on this line of work, subsequent methods focused on improving descriptor discriminability through metric learning. Such as, CosPlace [4] employs a cosine similarity-based loss to improve generalization, while MixVPR [1] enhances representations by mixing features from multiple network levels. In recent work, MegaLoc [3] proposed architectural refinements combined with contrastive learning to improve scalability. Nevertheless, retrieval-based VPR methods remain largely appearance-driven and tend to struggle in challenging scenarios, such as perceptual aliasing or severe viewpoint changes.

2.2. Re-ranking with Image Matching

To alleviate these limitations, many VPR pipelines adopt a second-stage geometric re-ranking step. In this stage, local features are matched between the query image and the top-ranked retrieval candidates, followed by geometric verification, typically using RANSAC. Modern deep matching approaches such as SuperGlue [8], LoFTR [9], and LightGlue [7], have substantially improved matching robustness, particularly in texture-poor or low-resolution environments. Nevertheless, these approaches are computationally demanding. Extracting and matching local features for each candidate significantly increases inference latency, leading to a fundamental trade-off between localization accuracy and computational efficiency.

2.3. Efficiency Aware VPR

Conventional VPR pipelines apply geometric verification to all queries in a uniform manner, regardless of their difficulty. This strategy is inherently inefficient, since a large portion of queries can already be localized correctly using global retrieval alone. Although conditional computation and early-exit strategies have been widely explored in general computer vision, their application to query-level decision-making in VPR remains limited. This observation highlights the need for efficiency-aware approaches that selectively trigger re-ranking only when global retrieval is unreliable. By allocating computational resources adaptively, such methods aim to preserve high localization accuracy while substantially reducing overall runtime.

3. Methodology

3.1. VPR pipeline overview

We adopt a standard two-stage Visual Place Recognition (VPR) pipeline consisting of global retrieval followed by optional geometric re-ranking. Given a query image q, the objective is to localize it by matching against a geo-referenced image database.

In the global retrieval stage, a global descriptor $d_q \in \mathbb{R}^D$ is extracted using a pre-trained VPR model. Descriptors $\{d_i\}_{i=1}^N$ for the database images are computed offline. A nearest-neighbor search is performed in the descriptor space to retrieve the top-K most similar images, yielding a ranked list of candidates $\{i_1, i_2, \ldots, i_K\}$, where $K = 20$ by default. The retrieval-only estimate corresponds to the top-ranked candidate i_1.

Despite the efficiency and scalability of global retrieval, its holistic nature often leads to failures under challenging conditions such as perceptual aliasing, viewpoint shifts, or strong appearance changes. To improve robustness, a geometric re-ranking stage can be applied to the retrieved candidates. In this stage, we match the query image against a subset of retrieved candidates using a local feature matching method, estimate geometric consistency with RANSAC, and compute an inlier count for each candidate. Candidates are then re-ordered by their inlier counts and the top-ranked candidate after re-ranking is taken as the final prediction.

Applying geometric re-ranking to every query generally improves accuracy but incurs substantial computational overhead. Motivated by this trade-off, we employ an adaptive framework that triggers re-ranking only for queries that are likely to be difficult, reducing the average computational cost while preserving most of the accuracy gains provided by geometric verification.

3.2. Query Difficulty Estimation

A central component of adaptive re-ranking is estimating whether a query is *easy*, for which retrieval-only is likely sufficient, or *hard*, for which geometric re-ranking is likely beneficial. We quantify query difficulty using the geometric consistency between the query and its top-1 retrieved candidate as an interpretable signal.

By considering a query image q and its top-1 retrieved database image i_1, we perform local feature matching fol-

lowed by geometric verification. Tentative feature correspondences are first established between q and i_1, after which RANSAC is applied to fit a geometric model and identify inlier correspondences, i.e., matches that are consistent with the estimated model. Let $n(q)$ denote the number of inliers obtained for the image pair (q, i_1).

Intuitively, a large inlier count indicates strong geometric agreement, suggesting that the top-1 retrieved image depicts the same physical location as the query. Conversely, a small inlier count implies weak geometric consistency and a higher likelihood that the top-1 retrieval is incorrect. We therefore treat $n(q)$ as a proxy for query difficulty, where queries with low inlier counts are considered harder and more likely to benefit from geometric re-ranking.

Importantly, this difficulty estimation relies solely on the top-1 retrieved candidate rather than all K retrieved candidates, which keeps the additional computational overhead minimal. We empirically validate the discriminative power of the inlier count by analyzing the distribution of $n(q)$ for correct and incorrect retrievals in Section 4.4.

3.3. Adaptive Re-ranking Strategies

Based on the difficulty estimate $n(q)$, we design an adaptive re-ranking mechanism that determines, for each query, whether to invoke the computationally expensive geometric re-ranking stage. We consider two alternative triggering strategies: a fixed hard threshold and a learned logistic trigger.

Hard Threshold Trigger. We define a threshold τ on the inlier count. If $n(q) < \tau$, the query is classified as hard and geometric re-ranking is applied to the retrieved candidate list $\{i_1, \ldots, i_K\}$. Otherwise, the query is treated as easy and the retrieval-only prediction i_1 is returned without further processing. Formally, the final prediction $\hat{i}(q)$ is given by

$$\hat{i}(q) = \begin{cases} \mathrm{ReRank}\big(q, \{i_1, \ldots, i_K\}\big), & \text{if } n(q) < \tau, \\ i_1, & \text{otherwise.} \end{cases} \quad (1)$$

This strategy is simple, interpretable, and directly controls the fraction of queries that trigger geometric re-ranking.

Logistic Regression Trigger. As a more flexible alternative to a fixed threshold, we propose a lightweight logistic regression–based trigger that learns a smooth decision boundary for adaptive re-ranking. The model takes as input a low-dimensional feature vector $\mathbf{x}_q$ constructed from signals readily available in the retrieval pipeline. In our implementation, $\mathbf{x}_q$ includes the inlier count $n(q)$ and a descriptor-based confidence measure given by the distance margin between the top-1 and top-2 retrieved candidates.

The logistic regression outputs a score $p(q) \in [0, 1]$, which reflects the likelihood that applying geometric re-ranking will be beneficial for the query. Geometric re-ranking is triggered when $p(q) > p_0$, where p_0 is a decision threshold selected on a validation set.

In our setting the logistic regression is used solely as a learned triggering mechanism for adaptive re-ranking, serving as a smooth, data-driven alternative to a fixed inlier threshold rather than as a standalone uncertainty estimator or failure predictor.

Both triggering strategies share the same downstream geometric re-ranking module and differ only in how the decision to invoke re-ranking is made. In Section 4.5, we compare the two triggers in terms of Recall@N and computational cost.

We include the logistic trigger primarily to assess whether a learned decision boundary can improve the performance–efficiency trade-off compared to a fixed inlier threshold, while remaining within the adaptive re-ranking framework.

3.4. Training and Validation Protocol

We select the triggering parameters using a validation protocol designed to balance localization performance and computational efficiency.

For the hard-threshold trigger, we evaluate a set of candidate inlier thresholds τ on a validation set. For each threshold, we measure both localization performance (e.g., Recall@N) and the corresponding re-ranking frequency. We then select the threshold that achieves the best trade-off between accuracy and computational cost.

For the logistic regression trigger, we construct training samples using queries from a held-out validation split of the SVOX dataset, including both the Sun and Night subsets. For each query q, we compute a low-dimensional feature vector $\mathbf{x}_q$ composed of two signals readily available in the retrieval pipeline: (i) the number of geometric inliers $n(q)$ obtained by matching the query with its top-1 retrieved candidate, and (ii) a descriptor-based confidence measure defined as the distance margin between the top-1 and top-2 retrieved candidates. And this margin is computed as the difference between their global descriptor distances to the query, i.e., $\Delta d = d(q, i_2) - d(q, i_1)$, where $d(\cdot, \cdot)$ denotes the L2 distance in descriptor space.

Each query is assigned a binary label indicating whether applying geometric re-ranking is beneficial. A query is labeled as positive if geometric re-ranking corrects an initially incorrect retrieval or otherwise improves localization correctness compared to retrieval-only, and negative otherwise.

We train the logistic regression model on these labeled samples and subsequently sweep the decision threshold p_0 on the validation set. The final operating point is selected to achieve a favorable balance between localization accuracy

and computational efficiency, consistent with the evaluation criteria used for the hard-threshold trigger.

All results reported on the test sets are obtained using parameters fixed after validation. This protocol prevents tuning on the test data and allows us to assess how well the adaptive triggering strategies generalize across datasets and difficulty levels.

4. Experiments and Results

4.1. Datasets

We evaluate our Visual Place Recognition (VPR) pipeline on Tokyo-XS [10], SF-XS [5], and SVOX, where SVOX [6] is reported separately on Day and Night subsets. Tokyo eXtra Small (Tokyo-XS): An urban dataset with moderate viewpoint and illumination changes, commonly used for fast prototyping and controlled evaluation. San Francisco Testing (SF-XS test): A challenging city-scale dataset featuring larger viewpoint changes, dynamic objects, and architectural ambiguity. SVOX (Day and Night subsets): A dataset specifically designed to evaluate robustness under severe appearance changes. We separately evaluate:SVOX-Day, which mainly contains viewpoint and seasonal variations. SVOX-Night, which is significantly more challenging due to extreme illumination changes, low texture visibility, motion blur, and sensor noise. The night subset represents one of the hardest test scenarios for global-descriptor-based VPR.

4.2. Evaluation Metrics

We report performance using the Recall@N (R@N) metric, which measures whether at least one correct match is retrieved within the top-N candidates. We evaluated R@1, R@5, R@10, R@20. This metric is standard in VPR and directly reflects the effectiveness of retrieval systems

4.3. Global Retrieval Models

We evaluate four state-of-the-art global descriptor methods:

- **NetVLAD** (4096-D) with a VGG-16 backbone (138 million parameters)
- **CosPlace** (512-D) with a ResNet-18 backbone (11 million parameters)
- **MixVPR** (4096-D) with a ResNet-50 backbone (25 million parameters)
- **MegaLoc** based on a DINOv2B backbone (87 million parameters)

For comparing query descriptors against the database, we evaluated both L2 distance and dot-product similarity. Since all global descriptors are L2-normalized, the two measures are mathematically equivalent and yielded identical retrieval results in practice. We adopt L2 distance for all subsequent experiments. Table 1 summarizes the experimental setup.

Item	Choice
Testsets	Tokyo-XS SF-XS SVOX-Day SVOX-Night
Models	NetVLAD CosPlace MixVPR MegaLoc
Metric	Recall@1/5/10/20
Top-K for re-ranking	K=20
Runtime reporting	Re-ranking time per query

Table 1. Experimental setup summary

4.4. Retrieval Only Baseline Results

We first evaluate retrieval-only performance, without any geometric re-ranking, to isolate the quality of the global descriptors.

The results of these experiments are highlighted in Table 2. From the results, across different datasets, we can make some conclusions:

- NetVLAD performs weakest, especially under strong viewpoint or illumination changes, reflecting the limitations of older CNN-based architecture despite the high number of parameters. The biggest difference is in the SVOX_Night dataset where Netvlad achieves 8.0 Recall@1 while MegaLoc achieves 95.1.
- CosPlace provides a noticeable improvement, benefiting from metric learning and a more efficient descriptor, but still struggles in more challenging conditions.
- MixVPR consistently outperforms both NetVLAD and CosPlace, particularly at higher recall levels (R@10, R@20), indicating better global context modeling.
- MegaLoc achieves the best performance across all datasets and recall thresholds, with especially strong gains on SF-XS and SVOX-Night, demonstrating the advantage of a modern transformer-based architecture. MegaLoc consistently achieves 85> R@1 across the different datasets.

The performance gap between methods widens as dataset difficulty increases, with SVOX-Night showing the largest relative differences. This highlights the importance of strong global representations when appearance changes like daylight dominate.

4.5. Re-ranking Performance and Cost

To further improve localization accuracy, we apply local-feature-based re-ranking on the top-N retrieved candidates. We evaluate three state-of-the-art feature matching

pipelines: SuperGlue, SuperPoint + LightGlue, LoFTR.

Table 2 highlights the performance of each method, while Table 3 highlights the time needed with each method to process the local-feature re-ranking for each query. In terms of time per query:

- SuperGlue is the fastest, making it suitable for real-time or resource-constrained systems. (1.47 seconds worst time)
- SuperPoint + LightGlue is slower offering better accuracy when compared to superglue.
- LoFTR is the slowest, compared to the others, offering better accuracy when compared to SuperGlue.

4.5.1. Performance Comparison

From a performance standpoint:

- SuperGlue consistently yields lower re-ranking accuracy, especially on difficult cases such as SVOX-Night. Highlighting a clear tradeoff between speed and accuracy. In SVOX-Night, Cosplace with SuperGlue achieve 64.5 R@5 compared to LoFTR and SuperPoint+LightGlue, respectivly 65.4 and 65.1.
- SuperPoint + LightGlue and LoFTR achieve similar levels of performance, and it is difficult to consistently rank one above the other across all datasets.
- Both outperform SuperGlue by a clear margin, particularly when illumination changes are severe.

Interestingly, for very strong global retrieval (MegaLoc), always re-ranking can sometimes degrade Recall@1, likely due to mismatches and geometric verification failures on hard image pairs. This motivates selective triggering: re-ranking should be applied only when it is expected to help. Overall, these results indicate that strong global retrieval combined with a robust but computationally efficient re-ranking stage offers the best performance–cost trade-off for practical VPR systems.Always re-ranking is expensive and may be unnecessary for strong retrievers, motivating selective triggering.

4.6. Correlation Between Retrieval Correctness and Number of Inliers

After the initial retrieval-only experiments, we analyze the relationship between retrieval correctness at R@1 and the number of geometric inliers obtained when matching the query image with its top-1 retrieved database image. For each query, we record:

- Whether the top-1 retrieved image is correct or incorrect (according to ground truth),
- The number of inliers obtained after local feature matching and geometric verification.

We then plot histograms of the number of inliers, separating queries into correct and wrong retrievals. Figure 1 highlight the inliner differences between correct and incorrect retrieved images.

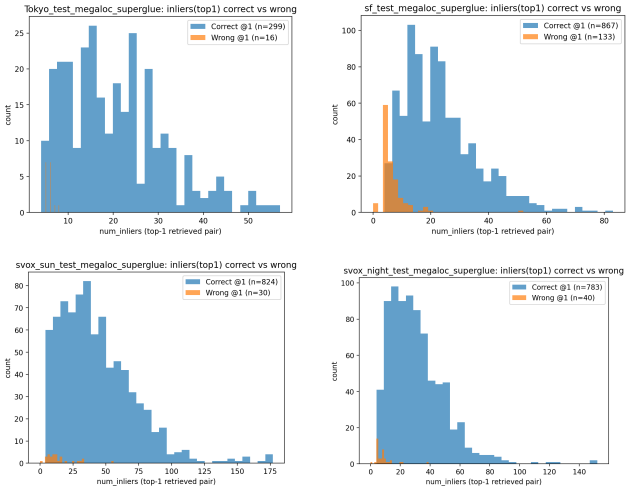

Figure 1. Histograms for number of inliners of correct and incorrect top-1 retrieved images.

4.6.1. Observations from Inlier Distributions

The histograms in Figure 1 reveal a clear statistical correlation between retrieval correctness and the number of inliers:

Correct queries tend to produce a significantly higher number of inliers, often forming a distribution with a long tail toward large inlier counts. Wrong queries are concentrated around low inlier counts, frequently close to zero or within a narrow low-inlier range.

This trend is consistent across datasets, although the absolute number of inliers varies depending on scene complexity and illumination conditions. On Tokyo-XS and SF-XS, correct matches typically yield a compact but clearly separable inlier distribution. On SVOX-Night, both distributions shift toward lower inlier counts due to reduced texture and poor lighting, but the relative separation between correct and incorrect queries remains observable.

While the number of inliers alone is not a perfect classifier, it provides a strong discriminative signal:

- Queries with very low inlier counts are overwhelmingly associated with incorrect retrievals.
- Queries with high inlier counts are almost always correct.
- An ambiguous overlap region exists at intermediate inlier values, where both correct and incorrect matches may occur.

This suggests that a simple threshold on the number of inliers could be used as a confidence measure for retrieval correctness. However, due to the overlap between the inlier distributions of correct and incorrect queries, such a threshold could inevitably introduce in some cases retrieval of false positives. Although rare, there exists the possibility that an incorrect candidate may yield a higher number of inliers than the correct top-1 retrieval, for example in the pres-

Method	Tokyo	SF	SVOX_Night	SVOX_Day
Cosplace	89.5	81.4	67.7	88.8
Base	65.1 / 79.7 / 86.0	63.1 / 74.8 / 78.6	33.3 / 51.5 / 59.1	62.3 / 78.5 / 84.5
+ SuperPoint+LightGlue	82.9 / 86.7 / 87.9	77.7 / 80.4 / 80.8	60.6 / 65.1 / 66.3	83.8 / 86.9 / 87.9
+ LoFTR	84.8 / 86.7 / 87.9	77.4 / 79.7 / 80.6	61.4 / 65.4 / 66.2	85.1 / 87.1 / 87.9
+ SuperGlue	82.5 / 86.3 / 88.6	76.7 / 79.4 / 80.5	59.4 / 64.5 / 65.9	81.3 / 85.9 / 87.4
NetVLAD	78.7	56.2	29.6	65.8
Base	49.8 / 62.5 / 70.5	27.2 / 43.8 / 50.3	8.0 / 17.4 / 23.1	35.4 / 52.7 / 58.8
+ SuperPoint+LightGlue	68.3 / 72.1 / 73.7	53.2 / 55.4 / 55.8	25.4 / 27.1 / 28.4	61.7 / 63.1 / 64.2
+ LoFTR	68.3 / 72.7 / 73.7	53.6 / 55.3 / 55.8	25.5 / 28.1 / 28.6	61.6 / 63.6 / 64.5
+ SuperGlue	69.5 / 72.7 / 74.9	52.4 / 55.3 / 55.8	24.5 / 26.6 / 28.1	60.4 / 62.9 / 64.2
MixVPR	93.7	83.9	88.0	95.9
Base	78.1 / 89.5 / 92.4	70.2 / 79.0 / 81.3	62.9 / 79.8 / 84.1	85.4 / 93.0 / 94.7
+ SuperPoint+LightGlue	89.2 / 92.4 / 92.7	81.4 / 83.3 / 83.7	81.9 / 86.3 / 87.1	91.7 / 95.0 / 95.6
+ LoFTR	89.8 / 92.4 / 93.7	80.1 / 82.9 / 83.7	82.5 / 86.8 / 87.6	93.4 / 95.0 / 95.3
+ SuperGlue	86.7 / 91.7 / 93.3	80.1 / 82.7 / 83.2	81.9 / 86.5 / 87.5	89.8 / 94.0 / 95.2
MegaLoc	99.0	91.3	99.1	99.6
Base	94.9 / 97.5 / 98.4	86.7 / 89.9 / 90.8	95.1 / 98.1 / 98.8	96.5 / 99.4 / 99.6
+ SuperPoint+LightGlue	94.3 / 98.4 / 99.0	86.8 / 90.2 / 90.7	90.8 / 97.4 / 98.8	95.8 / 99.3 / 99.5
+ LoFTR	94.9 / 97.5 / 98.4	85.6 / 89.3 / 90.6	92.3 / 97.9 / 98.8	96.7 / 99.4 / 99.5
+ SuperGlue	93.7 / 97.5 / 98.4	85.7 / 89.4 / 90.2	90.0 / 97.6 / 98.5	93.9 / 98.9 / 99.6

Table 2. Quantitative comparison on four datasets. For each method block, the **top row** (bold method name) indicates **Recall@20**. The rows below report **Recall@1 / Recall@5 / Recall@10** for the base method and its re-ranking variants.

Method	Time per Query (s)
Cosplace + SuperPoint+LightGlue	3.55
Cosplace + LoFTR	3.73
Cosplace + SuperGlue	1.44
NetVLAD + SuperPoint+LightGlue	3.52
NetVLAD + LoFTR	3.74
NetVLAD + SuperGlue	1.41
MixVPR + SuperPoint+LightGlue	3.52
MixVPR + LoFTR	3.69
MixVPR + SuperGlue	1.46
MegaLoc + SuperPoint+LightGlue	3.52
MegaLoc + LoFTR	3.70
MegaLoc + SuperGlue	1.47

Table 3. Average re-ranking time per query (in seconds) for various combinations of global retrieval models and local feature matching pipelines.

ence of repetitive structures or visually similar scenes. In such cases, a strict inlier threshold or aggressive re-ranking strategy could incorrectly discard a correct retrieval in favor of a visually consistent but semantically wrong match. In addition, relying on inlier-based verification adds an increased computational cost, as local feature matching and geometric verification are significantly more expensive than global descriptor retrieval.

4.7. Adaptive Re-ranking Results

Adaptive policy. Motivated by the observation in Section 4.4 that the top-1 inlier count correlates with retrieval correctness (Figure 1), we design an adaptive re-ranking policy that triggers expensive geometric verification only for "hard" queries. For each query, we compute the number of verified inliers between the query and its top-1 retrieved database image. If the inlier count is below a threshold t, we apply re-ranking over the top-K retrieved candidates; otherwise we keep the original retrieval ranking. In short, we re-rank if inlierstop1¡ t . We set K=20 in all experiments.

Validation sweep. We select the threshold t on the SF-XS validation split by sweeping a grid of candidate thresholds. For each t, we evaluate the resulting adaptive Recall@1 and the fraction of queries that trigger re-ranking. This sweep exposes a direct accuracy–compute trade-off: larger t re-ranks more queries (higher cost) and can improve recall, while smaller t saves computation but may miss hard cases. We choose t as the threshold that maximizes Recall@1 on validation; if multiple thresholds achieve the same best Recall@1, we select the one with the smallest re-ranking fraction. Based on this analysis on the SF-XS validation set, we identified $\tau = 6$ as the optimal threshold. Figure 2 highlight how the threshold choice influences

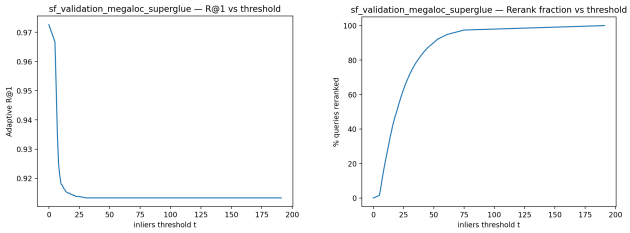

Figure 2. Adaptive re-ranking performance metrics: Recall@1 and re-ranking frequency as a function of the inlier threshold t on the validation split.

Recall@1 and the percentage of re-ranked queries.

Test performance. Using the threshold t selected on the SF-XS validation split for the MegaLoc + SuperGlue configuration, we evaluate three settings on each test set: (1) retrieval-only, (2) always re-ranking the top-20 candidates, and (3) adaptive re-ranking with the hard-threshold trigger. Across datasets, adaptive re-ranking preserves most of the Recall@K gains of always re-ranking, while avoiding unnecessary verification on easy queries where the top-1 retrieval is already reliable. As shown in Table 4, this allows us to achieve comparable recall performance while applying geometric re-ranking to only a small fraction of queries. Specifically, re-ranking is triggered for only 5.4%, 7.9%, 3.6%, and 1.5% of queries on Tokyo-XS, SF-XS, SVOX Night, and SVOX Day, respectively.

Trade-off and compute savings. To characterize the full operating range of the adaptive policy, we plot Recall@1 as a function of the re-ranking fraction (Figure 3), together with the corresponding time saved relative to always re-ranking. These curves, in Figures 2 and 4 illustrate that the method can be tuned to different compute budgets: operating points with modest re-ranking rates can retain most of the recall improvement while substantially reducing average runtime.

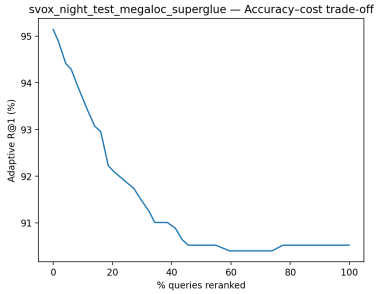

Figure 3. Relationship between localization accuracy (Recall@1) and computational cost (percentage of queries re-ranked) on the nighttime test split.

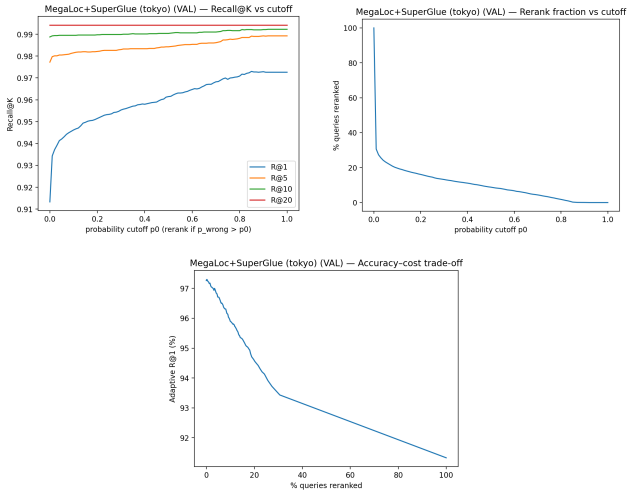

Figure 4. Analysis of the accuracy-cost trade-off. Relationship between Recall@1 and the percentage of queries triggering the re-ranking stage for the svox_night dataset and the Tokyo validation set.

Logistic trigger (ablation). As an additional ablation, we replace the fixed threshold with a lightweight logistic trigger and sweep the cutoff on validation. Under similar compute budgets, the learned trigger yields comparable trade-offs to the hard threshold, indicating that a simple inlier threshold already provides a strong and interpretable baseline. Figure 4 show how different levels of recall are affected by the cutoff probability.

As shown in Table 5, the logistic trigger achieves comparable Recall@1 to hard thresholding while re-ranking substantially fewer queries. This results in a lower average runtime per query, demonstrating that the learned trigger can identify a more cost-effective operating point without sacrificing localization accuracy. Unlike hard thresholding, the logistic trigger additionally provides a continuous confidence score, which can be analyzed using PR curves and AUPRC for uncertainty estimation.

5. conclusion

In this work, we investigated the accuracy–efficiency trade-off of a standard two-stage Visual Place Recognition pipeline, combining global descriptor retrieval with local geometric verification for re-ranking. Across multiple benchmark datasets, we observed a clear relationship between the number of geometric inliers computed for the top-1 retrieved candidate and the correctness of the initial retrieval, which provides an intuitive signal of query "hardness". Leveraging this signal, we designed an adaptive re-ranking strategy that selectively triggers expensive geometric verification only when the initial retrieval is likely to be unreliable. This adaptive mechanism aims to retain most of the Recall@N gains of always re-ranking while substan-

Dataset	Type	Threshold	R@1	R@5	R@10	Reranked	Time (s)
Tokyo	Retrieval	–	94.92	97.46	98.41	–	–
	Rerank	–	94.92	97.46	98.41	–	–
	Adaptive	6	95.24	98.10	98.73	5.4%	0.07
SF	Retrieval	–	86.70	89.90	90.80	–	–
	Rerank	–	85.90	89.40	90.40	–	–
	Adaptive	6	86.40	89.50	90.60	7.9%	0.10
SVOX_Night	Retrieval	–	95.14	98.06	98.78	–	–
	Rerank	–	90.52	97.45	98.42	–	–
	Adaptive	6	94.65	97.69	98.66	3.6%	0.04
SVOX_Sun	Retrieval	–	96.49	99.41	99.65	–	–
	Rerank	–	94.26	98.95	99.65	–	–
	Adaptive	6	95.90	99.30	99.65	1.5%	0.02

Table 4. Retrieval, reranking, and adaptive reranking performance across datasets

Trigger	R@1	Re-ranked (%)	Time / query (s)
Hard threshold	68.57	77.8	2.90
Logistic regression	68.75	59.7	2.22

Table 5. Comparison between hard-threshold and logistic triggers on Tokyo-XS using NetVLAD retrieval and LoFTR re-ranking.

tially reducing the average computational cost. We compared a simple hard inlier threshold with a lightweight logistic regression trigger, used strictly as a decision module for when to re-rank, and found that the learned trigger can match or slightly improve the performance–cost trade-off depending on the dataset and operating point. Overall, our results suggest that adaptive re-ranking is a practical way to make local verification scalable, especially in challenging conditions where always re-ranking is prohibitively expensive. A limitation of our approach is that the trigger depends on the quality of the top-1 local matching signal and may be sensitive to dataset shift (e.g., day vs. night).

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